

SPSS Assignment 1

Categorical Data Analysis: Frequency Distributions, Cross-Tabulation, Non-Parametric Tests, and Binary Logistic Regression

Datasets: Dataset 1 – Student_Academic_Performance.sav (N = 200); Dataset 2 – Employee_Job_Satisfaction.sav (N = 200)

Significance level: $\alpha = .05$ for all tests unless otherwise stated

Section A. Frequency Distribution & Visualization of Categorical Variables

Q1. Frequency Tables and Bar Charts – Department, Study_Method, Scholarship (Dataset 1)

SPSS Steps

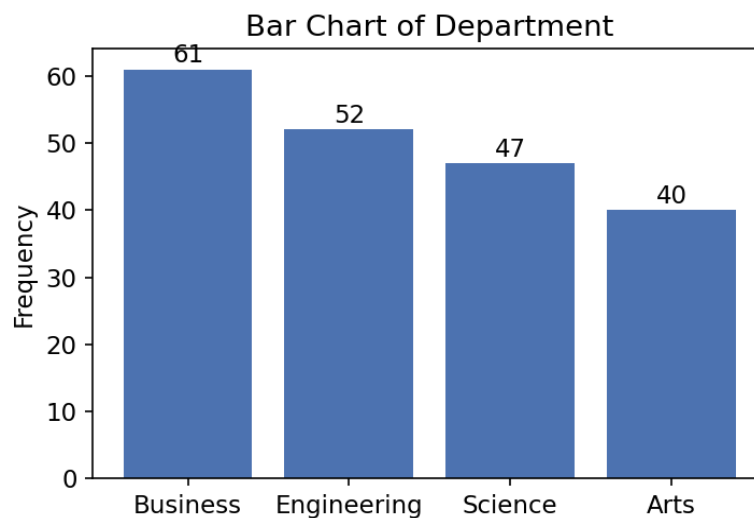
- Analyze → Descriptive Statistics → Frequencies
- Move Department, Study_Method, and Scholarship into the Variable(s) box
- Click Charts → select Bar charts → Frequencies → Continue
- Ensure “Display frequency tables” is checked → OK

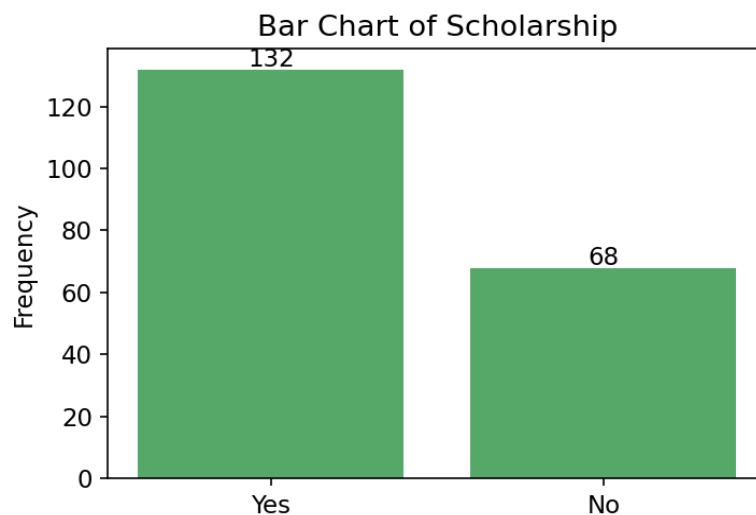
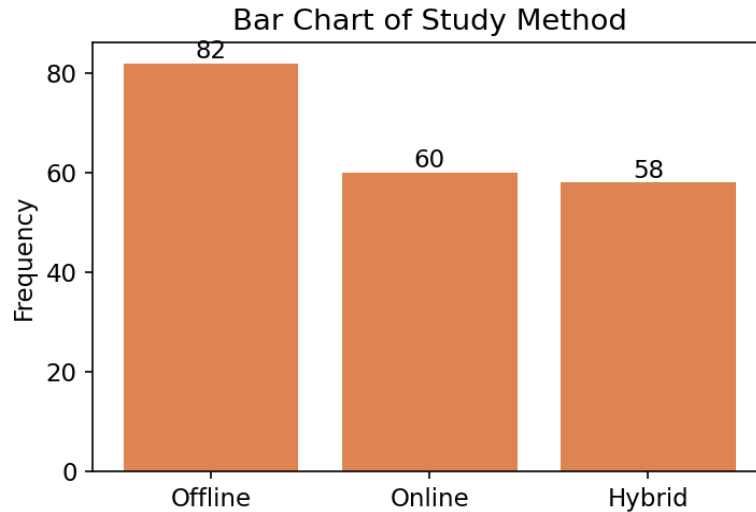
Output

Department	n	%
Business	61	30.5
Engineering	52	26.0
Science	47	23.5
Arts	40	20.0
Total	200	100.0

Study Method	n	%
Offline	82	41.0
Online	60	30.0
Hybrid	58	29.0
Total	200	100.0

Scholarship	n	%
Yes	132	66.0
No	68	34.0
Total	200	100.0





APA-Style Interpretation

Of the 200 students sampled, the largest proportion were enrolled in the Business department ($n = 61$, 30.5%), followed by Engineering ($n = 52$, 26.0%), Science ($n = 47$, 23.5%), and Arts ($n = 40$, 20.0%). Regarding study method, most students studied Offline ($n = 82$, 41.0%), while Online ($n = 60$, 30.0%) and Hybrid ($n = 58$, 29.0%) methods were similarly common. The majority of students had received a scholarship ($n = 132$, 66.0%), compared to 34.0% ($n = 68$) who had not.

Q2. Frequency Tables and Pie Charts – Education_Level, Marital_Status, Turnover_Intention (Dataset 2)

SPSS Steps

- Analyze → Descriptive Statistics → Frequencies
- Move Education_Level, Marital_Status, and Turnover_Intention into the Variable(s) box
- Click Charts → select Pie charts → Frequencies → Continue → OK

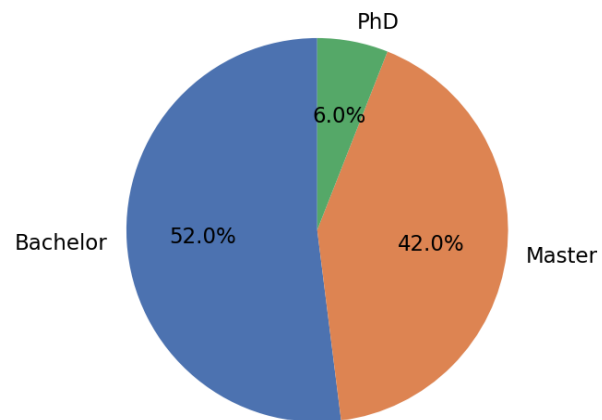
Output

Education Level	n	%
Bachelor	104	52.0
Master	84	42.0
PhD	12	6.0
Total	200	100.0

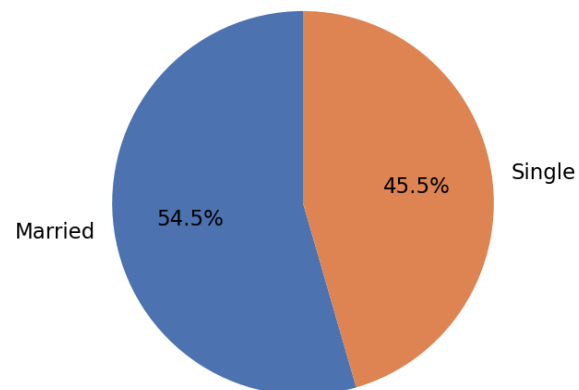
Marital Status	n	%
Married	109	54.5
Single	91	45.5
Total	200	100.0

Turnover Intention	n	%
No	153	76.5
Yes	47	23.5
Total	200	100.0

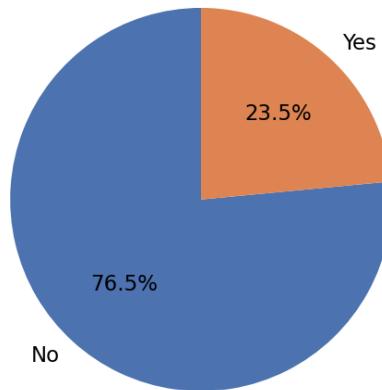
Pie Chart of Education Level



Pie Chart of Marital Status



Pie Chart of Turnover Intention



APA-Style Interpretation

Among the 200 employees, over half held a Bachelor's degree ($n = 104$, 52.0%), 42.0% ($n = 84$) held a Master's degree, and 6.0% ($n = 12$) held a PhD. Slightly more employees were married ($n = 109$, 54.5%) than single ($n = 91$, 45.5%). Most employees reported no intention to leave the organization within the next year ($n = 153$, 76.5%), whereas 23.5% ($n = 47$) indicated an intention to leave (turnover intention).

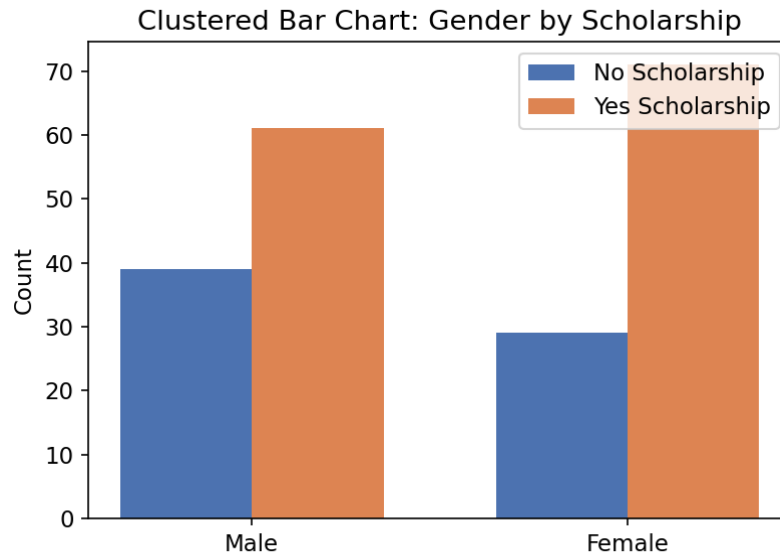
Q3. Clustered Bar Charts – Gender $\times$ Scholarship (Dataset 1) and Department $\times$ Turnover_Intention (Dataset 2)

SPSS Steps

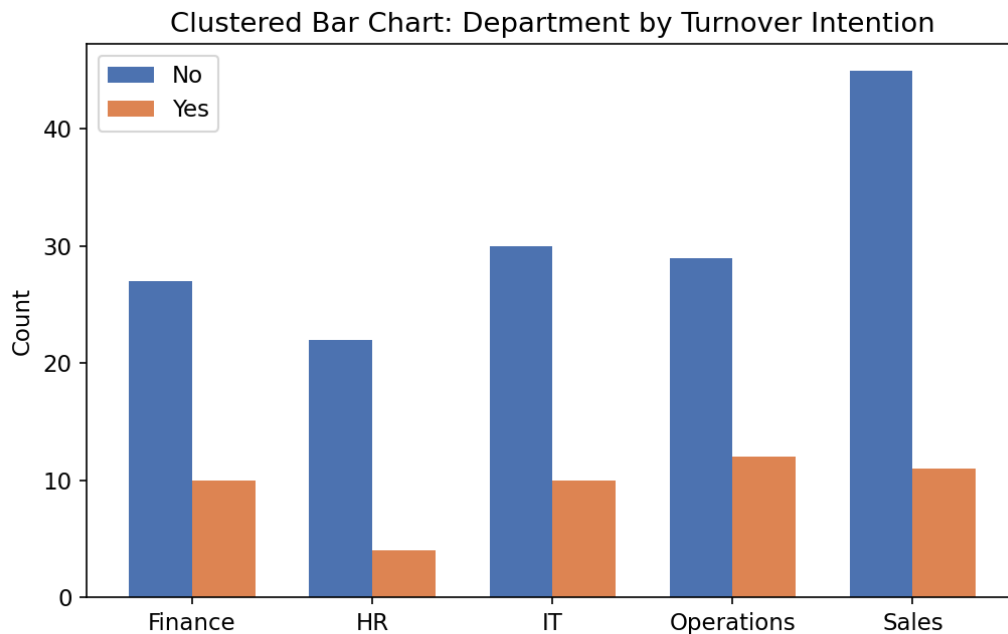
- Graphs $\rightarrow$ Chart Builder $\rightarrow$ choose Bar $\rightarrow$ Clustered Bar
- Dataset 1: drag Gender to the X-axis and Scholarship to the Cluster (color) box $\rightarrow$ OK
- Dataset 2: drag Department to the X-axis and Turnover_Intention to the Cluster (color) box $\rightarrow$ OK

Output

Gender	No	Yes	Total
Male	39	61	100
Female	29	71	100
Total	68	132	200



Department	No	Yes	Total
Finance	27	10	37
HR	22	4	26
IT	30	10	40
Operations	29	12	41
Sales	45	11	56
Total	153	47	200



APA-Style Interpretation

The clustered bar chart shows that a higher proportion of female students received a scholarship (71/100, 71.0%) compared to male students (61/100, 61.0%), although this difference is formally tested for

significance in Section C (Q6). For turnover intention, the proportion of employees intending to leave was fairly similar across departments, ranging from 15.4% in HR to 29.3% in Operations, with Sales showing the largest number of “No” responses in absolute terms.

Section B. Recoding Continuous Variables into Categorical Variables

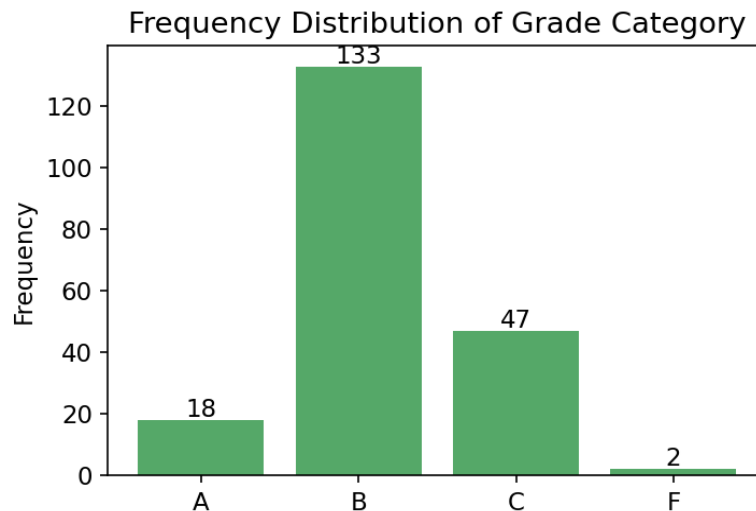
Q4. Recode Final_Exam_Score into Grade_Category (Dataset 1)

SPSS Steps

- Transform → Recode into Different Variables
- Move Final_Exam_Score into the Input Variable box; name the output variable Grade_Category
- Click Old and New Values: 0 thru 39.99 → 4 (F); 40 thru 59.99 → 3 (C); 60 thru 79.99 → 2 (B); 80 thru Highest → 1 (A)
- Assign value labels 1 = A, 2 = B, 3 = C, 4 = F → Change → OK
- Analyze → Descriptive Statistics → Frequencies → Grade_Category → OK

Output

Grade Category	n	%	Score Range Observed
A	18	9.0	80.3 – 99.5
B	133	66.5	60.1 – 79.9
C	47	23.5	44.2 – 59.8
F	2	1.0	5.7 – 12.1
Total	200	100.0	



APA-Style Interpretation

Recoding Final_Exam_Score using the specified cut-points ($A \geq 80$, $B = 60\text{--}79$, $C = 40\text{--}59$, $F < 40$) produced a right-skewed distribution of grades: the majority of students earned a grade of B ($n = 133$, 66.5%), followed by C ($n = 47$, 23.5%), A ($n = 18$, 9.0%), and F ($n = 2$, 1.0%). This indicates that most students performed in the moderate-to-good range on the final exam, with very few outright failures.

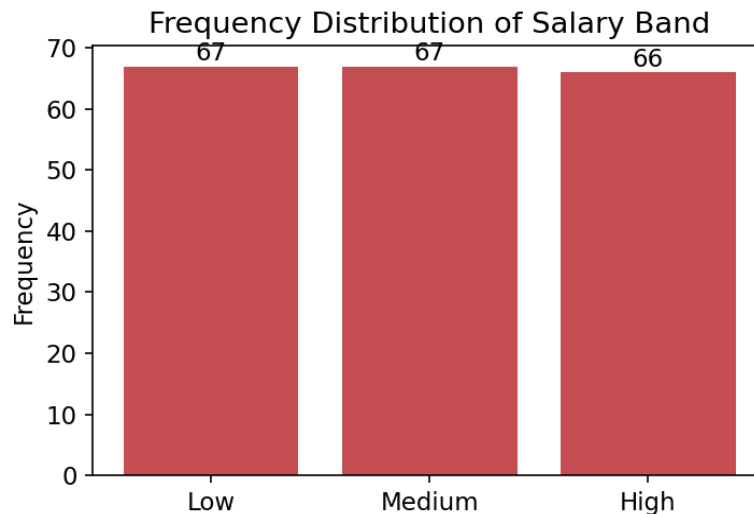
Q5. Recode Monthly_Salary into Salary_Band (Dataset 2)

SPSS Steps

- Analyze → Descriptive Statistics → Frequencies → Monthly_Salary → Statistics → Percentile values: 33.33, 66.67 → obtain tertile cut-points ($\approx$ BDT 48,964 and $\approx$ BDT 61,407)
- Transform → Recode into Different Variables → Input variable: Monthly_Salary; Output variable: Salary_Band
- Old and New Values: Lowest thru 48963.99 → 3 (Low); 48964 thru 61406.99 → 2 (Medium); 61407 thru Highest → 1 (High)
- Assign value labels 1 = High, 2 = Medium, 3 = Low → Change → OK
- Analyze → Descriptive Statistics → Frequencies → Salary_Band → OK

Output

Salary Band	n	%	Salary Range (BDT)	Mean (BDT)
Low	67	33.5	24,462 – 48,751	41,457
Medium	67	33.5	49,390 – 61,534	54,800
High	66	33.0	61,544 – 93,001	70,112
Total	200	100.0		



APA-Style Interpretation

Using tertile cut-points (approximately BDT 48,964 and BDT 61,407), Monthly_Salary was recoded into three approximately equal-sized bands: Low ($n = 67$, 33.5%; $M = \text{BDT } 41,457$), Medium ($n = 67$, 33.5%; $M = \text{BDT } 54,800$), and High ($n = 66$, 33.0%; $M = \text{BDT } 70,112$). Because tertiles were used, the three groups are, by construction, almost identical in size, which is useful for subsequent group comparisons.

Section C. Cross-Tabulation & Chi-Square Test of Independence

Q6. Gender $\times$ Scholarship (Dataset 1)

SPSS Steps

- Analyze → Descriptive Statistics → Crosstabs
- Row(s): Gender; Column(s): Scholarship
- Statistics → check Chi-square and Phi and Cramer's V → Continue

- Cells → check Observed and Expected counts → OK

H₀ (null): There is no association between Gender and Scholarship status (the variables are independent).

H₁ (alternative): There is an association between Gender and Scholarship status (the variables are not independent).

Output – Crosstabulation

Gender	No (Observed)	No (Expected)	Yes (Observed)	Yes (Expected)	Total
Male	39	34.0	61	66.0	100
Female	29	34.0	71	66.0	100
Total	68	68.0	132	132.0	200

Chi-Square Tests

Test	Value	df	p (2-sided)
Pearson Chi-Square	1.805	1	.179
Continuity Correction ^a	1.436	1	.231

^a Applied because the table is 2×2.

Assumption Check

All expected cell counts equal 34.0 or 66.0, well above the minimum threshold of 5; therefore 0% of cells have an expected count below 5, satisfying the chi-square assumption.

Measure of Association

Statistic	Value
Cramer's V	.095
p	.179

Result: A chi-square test of independence showed no statistically significant association between gender and scholarship status, $\chi^2(1, N = 200) = 1.81$, $p = .179$, Cramer's $V = .095$. The effect size is very small (V close to 0), indicating gender and scholarship receipt are essentially unrelated in this sample.

Decision: Fail to reject H₀ ($p > .05$).

Q7. Department × Turnover_Intention (Dataset 2)

SPSS Steps

- Analyze → Descriptive Statistics → Crosstabs
- Row(s): Department; Column(s): Turnover_Intention
- Statistics → Chi-square, Cramer's V → Continue; Cells → Observed & Expected → OK

H₀ (null): There is no association between Department and Turnover Intention.

H₁ (alternative): There is an association between Department and Turnover Intention.

Output – Crosstabulation (Observed / Expected)

Department	No (Obs.)	No (Exp.)	Yes (Obs.)	Yes (Exp.)	Total
Finance	27	28.30	10	8.70	37
HR	22	19.89	4	6.11	26
IT	30	30.60	10	9.40	40
Operations	29	31.36	12	9.64	41
Sales	45	42.84	11	13.16	56
Total	153	153.0	47	47.0	200

Chi-Square Test

Test	Value	df	p (2-sided)
Pearson Chi-Square	2.481	4	.648

Assumption Check

The smallest expected count is 6.11 (HR – Yes), and no cell has an expected count below 5 (0% of cells < 5), so the minimum-expected-count assumption for chi-square is satisfied.

Statistic	Value
Cramer's V	.111
p	.648

Result: A chi-square test of independence indicated no statistically significant association between department and turnover intention, $\chi^2(4, N = 200) = 2.48$, $p = .648$, Cramer's V = .111 (small effect).

Decision: Fail to reject H_0 ($p > .05$).

Q8. Study_Method × Extracurricular (Dataset 1)

SPSS Steps

- Analyze → Descriptive Statistics → Crosstabs
- Row(s): Study_Method; Column(s): Extracurricular
- Statistics → Chi-square, Cramer's V → Continue → OK

H_0 (null): There is no association between Study Method and participation in Extracurricular activities.

H_1 (alternative): There is an association between Study Method and participation in Extracurricular activities.

Output – Crosstabulation (Observed / Expected)

Study Method	No (Obs.)	No (Exp.)	Yes (Obs.)	Yes (Exp.)	Total
Hybrid	35	33.64	23	24.36	58
Offline	44	47.56	38	34.44	82
Online	37	34.80	23	25.20	60
Total	116	116.0	84	84.0	200

Chi-Square Test

Test	Value	df	p (2-sided)
Pearson Chi-Square	1.097	2	.578

Assumption Check

All expected counts (24.36–47.56) exceed 5, so 0% of cells have expected counts below 5 — the assumption is satisfied.

Statistic	Value
Cramer's V	.074
p	.578

Result: There was no statistically significant association between study method and extracurricular participation, $\chi^2(2, N = 200) = 1.10$, $p = .578$, Cramer's V = .074 (negligible effect).

Decision: Fail to reject H_0 ($p > .05$).

Section D. Non-Parametric Tests Involving Categorical Grouping Variables

Q9. Mann–Whitney U Test: Final_Exam_Score by Gender (Dataset 1)

SPSS Steps

- Analyze → Nonparametric Tests → Legacy Dialogs → 2 Independent Samples
- Test Variable List: Final_Exam_Score; Grouping Variable: Gender (define groups 1, 2)
- Test Type: Mann-Whitney U → OK
- (For comparison) Analyze → Compare Means → Independent-Samples T Test: Test Variable Final_Exam_Score, Grouping Variable Gender

H₀ (null): The distribution of Final_Exam_Score is the same across Gender groups.

H₁ (alternative): The distribution of Final_Exam_Score differs across Gender groups.

Output – Ranks

Gender	n	Mean Rank	Sum of Ranks
Male	100	95.54	9553.50
Female	100	105.47	10546.50

Test Statistics

Mann-Whitney U	Z	Asymp. Sig. (2-tailed)
4503.50	–1.211	.226

Result: A Mann-Whitney U test indicated that Final_Exam_Score did not differ significantly between male (Mdn rank = 95.54) and female (Mdn rank = 105.47) students, $U = 4503.50$, $Z = -1.21$, $p = .226$.

Comparison with the Parametric Result

An independent-samples t-test on the same variable produced a consistent conclusion: Levene's test showed equal variances could be assumed, $F = 0.85$, $p = .357$; the t-test was not significant, $t(198) = -1.54$, $p = .126$ (Male $M = 65.83$, $SD = 12.43$; Female $M = 68.24$, $SD = 9.65$). Both the non-parametric Mann-Whitney U test and the parametric independent-samples t-test therefore lead to the same conclusion — no statistically significant gender difference in final exam performance — which reinforces confidence in the result even though Final_Exam_Score was reasonably normally distributed.

Decision: Fail to reject H_0 ($p > .05$).

Q10. Kruskal–Wallis H Test: Performance_Rating across Department (Dataset 2)

SPSS Steps

- Analyze → Nonparametric Tests → Legacy Dialogs → K Independent Samples
- Test Variable List: Performance_Rating; Grouping Variable: Department (range 1–5)
- Test Type: Kruskal-Wallis H → OK

H₀ (null): The distribution of Performance_Rating is the same across all Department categories.

H₁ (alternative): The distribution of Performance_Rating differs across at least one Department category.

Output – Ranks

Department	n	Mean Rank
Finance	37	87.88

Sales	56	107.79
IT	40	87.68
Operations	41	104.07
HR	26	116.87
Total	200	

Test Statistics

Kruskal-Wallis H	df	Asymp. Sig.
6.846	4	.144

Result: A Kruskal-Wallis H test showed that supervisor-rated Performance_Rating did not differ significantly across the five departments, $H(4, N = 200) = 6.85$, $p = .144$. Mean ranks ranged from 87.68 (IT) to 116.87 (HR), but this range was not large enough to be statistically significant.

Decision: Fail to reject H_0 ($p > .05$).

Q11. Wilcoxon Signed-Rank Test: Q1_Engagement vs. Q2_Engagement (Dataset 2)

SPSS Steps

- Analyze → Nonparametric Tests → Legacy Dialogs → 2 Related Samples
- Pair the variables Q1_Engagement and Q2_Engagement
- Test Type: Wilcoxon → OK

H_0 (null): There is no difference in the distribution of engagement scores between Quarter 1 and Quarter 2 (median difference = 0).

H_1 (alternative): There is a difference in the distribution of engagement scores between Quarter 1 and Quarter 2.

Output – Ranks

	n	Mean Rank	Sum of Ranks
Negative Ranks (Q2 < Q1)	51	77.01	3927.50
Positive Ranks (Q2 > Q1)	148	107.92	15972.50
Ties (Q2 = Q1)	1		
Total	200		

Test Statistics

Z	Asymp. Sig. (2-tailed)
-7.404	< .001

Result: A Wilcoxon signed-rank test indicated that Q2 engagement scores ($M = 56.69$) were significantly higher than Q1 engagement scores ($M = 54.40$), $Z = -7.40$, $p < .001$. Of the 199 non-tied pairs, 148 (74.4%) showed an increase from Q1 to Q2, while only 51 (25.6%) showed a decrease, suggesting a genuine upward shift in employee engagement between the first two quarters.

Decision: Reject H_0 ($p < .05$).

Section E. Binary Logistic Regression

Q12. Scholarship (DV) on Final_Exam_Score and Study_Hours_Per_Week (Dataset 1)

SPSS Steps

- Analyze → Regression → Binary Logistic

- Dependent: Scholarship (coded 0 = No, 1 = Yes)
- Covariates: Final_Exam_Score, Study_Hours_Per_Week; Method: Enter
- Options → check Classification plots, Hosmer-Lemeshow, CI for Exp(B) → Continue → OK

H₀ (null): Final_Exam_Score and Study_Hours_Per_Week do not predict the odds of receiving a Scholarship (all $\beta = 0$).

H₁ (alternative): At least one of Final_Exam_Score and Study_Hours_Per_Week predicts the odds of receiving a Scholarship (at least one $\beta \neq 0$).

Model Fit

Statistic	Value
–2 Log Likelihood (Block 0, constant only)	256.41
–2 Log Likelihood (Block 1, final model)	226.15
Model χ^2 (Omnibus)	30.26
df	2
p	< .001
Cox & Snell R ²	.140
Nagelkerke R ²	.194

Variables in the Equation

Predictor	B	SE	Wald/z	p	Exp(B)	95% CI for Exp(B)
Final_Exam_Score	.085	.021	4.12	< .001	1.089	[1.046, 1.134]
Study_Hours_Per_Week	.044	.048	0.93	.351	1.045	[0.952, 1.148]
Constant	–5.477	1.311	–4.18	< .001	0.004	

Classification Table (cut-off = .50)

Observed	Predicted: No	Predicted: Yes	% Correct
No	27	41	39.7
Yes	14	118	89.4
Overall %			72.5

APA-Style Interpretation

A logistic regression was conducted to predict scholarship receipt from final exam score and weekly study hours. The overall model was statistically significant, $\chi^2(2, N = 200) = 30.26$, $p < .001$, and explained between 14.0% (Cox & Snell R²) and 19.4% (Nagelkerke R²) of the variance in scholarship status, correctly classifying 72.5% of cases. Final_Exam_Score was a significant positive predictor, $B = 0.085$, Wald $z = 4.12$, $p < .001$, $\text{Exp}(B) = 1.089$, 95% CI [1.046, 1.134] — each additional point on the final exam was associated with an 8.9% increase in the odds of receiving a scholarship.

Study_Hours_Per_Week was not a significant predictor once exam score was controlled for, $B = 0.044$, $p = .351$, $\text{Exp}(B) = 1.045$. The model was considerably better at correctly identifying scholarship recipients (89.4% sensitivity) than non-recipients (39.7% specificity).

Decision: Reject H₀ for the overall model and for Final_Exam_Score ($p < .05$); fail to reject H₀ for Study_Hours_Per_Week ($p > .05$).

Q13. Turnover_Intention (DV) on Monthly_Salary, Performance_Rating, and Overall Job Satisfaction (Dataset 2)

SPSS Steps

- Analyze → Regression → Binary Logistic

- Dependent: Turnover_Intention (0 = No, 1 = Yes)
- Covariates: Monthly_Salary, Performance_Rating, Overall_Job_Satisfaction (mean of the 12 job-satisfaction items); Method: Enter
- Options → CI for Exp(B), classification plot → Continue → OK

H₀ (null): Monthly_Salary, Performance_Rating, and Overall Job Satisfaction do not predict the odds of Turnover Intention (all $\beta = 0$).

H₁ (alternative): At least one of Monthly_Salary, Performance_Rating, and Overall Job Satisfaction predicts the odds of Turnover Intention (at least one $\beta \neq 0$).

Model Fit

Statistic	Value
–2 Log Likelihood (Block 0, constant only)	218.10
–2 Log Likelihood (Block 1, final model)	195.47
Model χ^2 (Omnibus)	22.63
df	3
p	< .001
Cox & Snell R ²	.107
Nagelkerke R ²	.161

Variables in the Equation

Predictor	B	SE	Wald/z	p	Exp(B)	95% CI for Exp(B)
Monthly_Salary	< .001	.00001	0.65	.516	1.000	[1.0000, 1.0000]
Performance_Rating	–.016	.021	–0.78	.438	.984	[0.945, 1.025]
Overall_Job_Satisfaction	–1.345	.371	–3.62	< .001	.261	[0.126, 0.539]
Constant	3.200	1.397	2.29	.022	24.537	

Classification Table (cut-off = .50)

Observed	Predicted: No	Predicted: Yes	% Correct
No	150	3	98.0
Yes	41	6	12.8
Overall %			78.0

APA-Style Interpretation

The overall model significantly predicted turnover intention, $\chi^2(3, N = 200) = 22.63, p < .001$, explaining 10.7% (Cox & Snell) to 16.1% (Nagelkerke) of the variance, and correctly classifying 78.0% of employees overall. Overall job satisfaction was the only significant predictor, $B = -1.345$, Wald $z = -3.62, p < .001$, $\text{Exp}(B) = 0.261$, 95% CI [0.126, 0.539] — for each one-unit increase in mean job satisfaction (on the 1–5 scale), the odds of intending to leave decreased by roughly 74%. Monthly salary ($p = .516$) and supervisor-rated performance ($p = .438$) were not significant predictors once job satisfaction was in the model. Despite acceptable overall accuracy, the model had very low sensitivity for detecting employees who intend to leave (12.8%), reflecting the class imbalance in the outcome (76.5% “No” vs. 23.5% “Yes”).

Decision: Reject H₀ for the overall model and for Overall_Job_Satisfaction ($p < .05$); fail to reject H₀ for Monthly_Salary and Performance_Rating ($p > .05$).

Section F. Reporting Categorical Association Summary

Across both datasets, purely demographic or categorical grouping variables were consistently the weakest correlates of the outcomes of interest, while continuous, performance- or attitude-based variables carried

most of the predictive signal. For Dataset 1, Gender showed only a small, non-significant association with Scholarship status in the chi-square test, $\chi^2(1, N = 200) = 1.81$, $p = .179$, Cramer's $V = .095$, indicating that being male or female was essentially unrelated to receiving a scholarship. In contrast, the binary logistic regression (Q12) identified Final_Exam_Score as a significant positive predictor of scholarship receipt, $\text{Exp}(B) = 1.089$, 95% CI [1.046, 1.134], $p < .001$, while Study_Hours_Per_Week was not significant ($p = .351$); the model explained a modest 19.4% of the variance (Nagelkerke R^2).

For Dataset 2, Department showed a similarly weak, non-significant association with Turnover_Intention, $\chi^2(4, N = 200) = 2.48$, $p = .648$, Cramer's $V = .111$, meaning which department an employee worked in did not meaningfully predict their intention to leave. The logistic regression (Q13) told a more informative story: Overall_Job_Satisfaction emerged as the only significant predictor of turnover intention, $\text{Exp}(B) = 0.261$, 95% CI [0.126, 0.539], $p < .001$, while Monthly_Salary ($p = .516$) and Performance_Rating ($p = .438$) were not significant; the model accounted for 16.1% of the variance (Nagelkerke R^2).

Taken together, these results suggest that in both the academic and organizational contexts examined here, broad demographic categories (gender, department) are poor predictors of scholarship or turnover outcomes, whereas variables that more directly capture performance (exam score) or attitudes (job satisfaction) are the stronger, statistically significant drivers. This pattern implies that interventions aimed at influencing scholarship allocation or reducing turnover would likely be more effective if they targeted academic performance and employee satisfaction directly, rather than relying on demographic profiling.